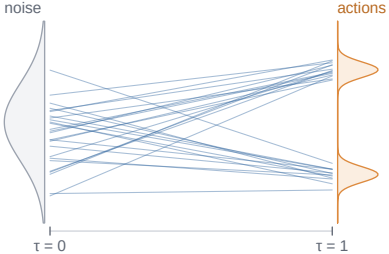
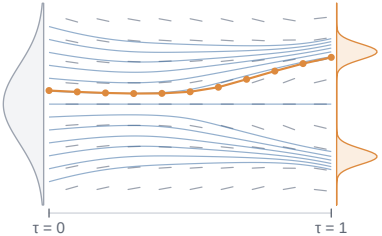
(a) Flow matching: learn a velocity field, integrate it from noise to action



training pairs (ε, A): straight paths cross



learned field: curved paths; Euler, N = 10 (dots)

training

$x_\tau = (1-\tau)\,\varepsilon + \tau A, \quad \varepsilon \sim N(0, I)$

$L(\theta) = E\,\|v_\theta(x_\tau, \tau \mid o) - (A - \varepsilon)\|^2$

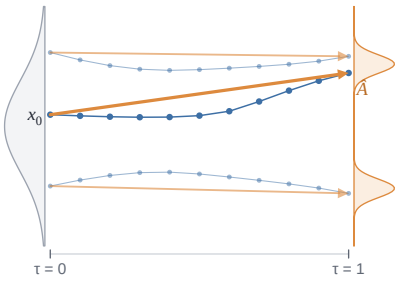
sampling

$x_0 = \varepsilon, \quad x_{\tau+\delta} = x_\tau + \delta\, v_\theta(x_\tau, \tau \mid o)$

δ = 1/N, one network call per step

paths of the marginal field are curved:
few large steps leave the path

(b) Distillation: N steps → 1 step



teacher: N = 10 Euler steps (frozen)

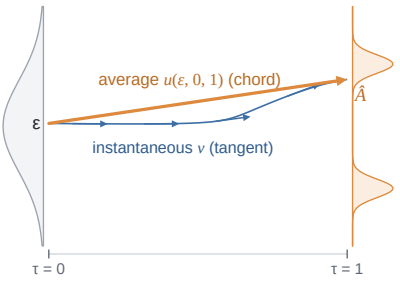
student S_φ: one jump, 1 network call

$L(\varphi) = \|S_\varphi(x_0, o) - \Phi^{(N)}_\theta(x_0, o)\|^2$

$\Phi^{(N)}_\theta$: N-step solve of the teacher field

NFE: N → 1; needs a trained teacher

(c) One-step objective: mean velocity



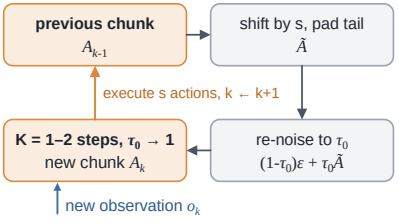
average velocity over [τ, τ']:

$(\tau'-\tau)\,u(x_\tau, \tau, \tau') = \int_\tau^{\tau'} v(x_s, s)\,ds$

one step: $\hat{A} = \varepsilon + u_\theta(\varepsilon, 0, 1 \mid o)$

u_θ trained directly on the average velocity
(no teacher; NFE = 1)

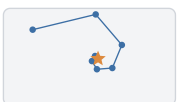
(d) Warm start and cross-step iteration



T_o = shift → re-noise → denoise:

$\|T_o(A) - T_o(A')\| \le \rho\|A - A'\|, \quad \rho < 1$

holds only while the observation o is fixed


fixed o: converges to $A^*(o)$


o_k changes: $A^*(o_k)$ moves